

Evaluate the following integrals:

1. $\int \frac{2}{x-3} dx$

2. $\int \frac{2}{x-3} + \frac{5}{(x-3)^2} - \frac{1}{(x-3)^3} dx$

3. $\int \cos^2 x \sin x dx$

4. $\int \frac{5}{\sqrt{x^2+4}} dx$

5. $\int x e^{2x} dx$

6. $\int x e^{x^2} dx$

7. $\int \sin^2(3x) dx$

8. $\int \frac{3}{x^2 + 4x} dx$

9. $\int 2x \cos(3x) dx$

10. $\int \sqrt{5 - x^2} \, dx$

11. $\int \frac{x}{x^2 - 25} \, dx$

12. $\int e^{2x} \sin(5x) \, dx$

13. $\int \frac{2x}{x^2 + 6x + 5} \, dx$

14. $\int \frac{x-1}{x^2+6x+9} dx$

15. $\int x^2 \sin(2x) dx$

16. $\int \frac{\sin(\ln 3x)}{2x} dx$

17. $\int \frac{4x + 2}{x^2 - x} dx$

18. $\int \frac{6x^2 - 7x}{3x - 2} dx$

19. $\int \frac{x^2 + 4x - 23}{(x^2 + 4)(x + 3)} dx$